GlobalProtect Connect Before Logon (PLAP) for Hybrid Azure AD Join Autopilot via Microsoft Intune — Deployment, Registration & Reboot-Sequencing Troubleshooting

Implementation Solution Guide

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Executive Summary — GlobalProtect Connect Before Logon (PLAP) for Hybrid Azure AD Join Autopilot via Microsoft Intune

The business problem

After a security incident, the organization moved off **Always On VPN** to a **user-initiated VPN** — but it still provisions Windows with **Hybrid Azure AD Join Windows Autopilot**, which needs the device to reach a **domain controller before the first user signs in** (for domain join, Group Policy, and a first-time logon with no cached credentials). A user-initiated VPN breaks that pre-logon reachability unless the tunnel can be raised **at the sign-in screen**. In practice the base **GlobalProtect** client deploys fine from Intune, but the **Connect Before Logon (PLAP)** registration is fragile: it intermittently fails with an **"unzipping" error** during Autopilot, reboot orchestration compounds it, and the **Connect Before Logon control is not reliably visible** at sign-in. Testing was on Windows 10; production is **Windows 11**. The rollout stalls, and every failure is hard to diagnose.

The solution

This is a complete, self-guided package that makes GlobalProtect Connect Before Logon **deployable AND diagnosable** for the hybrid-join Autopilot scenario. It delivers Connect Before Logon as a **single, correctly sequenced Intune Win32 app** — install the GlobalProtect MSI (5.2+), write the registry that enables Connect Before Logon and sets the portal, wait for the GlobalProtect service, register PLAP in the SYSTEM context, and **return exit code 3010 so the Enrollment Status Page owns the reboot** (never a scripted reboot). It makes the app a **blocking app** in the ESP, targets an **Assigned** Autopilot device group **owned by the Intune Provisioning Client service principal** so it populates at enrollment, and neutralizes the **"unzipping" 0x87D30067** error with an antivirus/IMECache exclusion and a clean re-wrap. A **log-capture runbook** turns any failed provisioning into a diagnosed cause on a Windows 11 test device.

What you get

- **Step-by-step implementation** at click-by-click detail on the Intune/Autopilot/Entra/Windows side —

the single Win32 app, the ESP blocking app, the enrollment-time device group, and the AV exclusion, with explicit Gate A–E pass/fail gates.

- **Five PowerShell tools** — readiness discovery (incl. the group's service-principal owner check),

on-device Windows 11 validation, a **log-capture collector** (IME + Autopilot + GlobalProtect logs), fleet coverage reporting, and an enrollment-time device-group builder that sets the SP owner.

- **Requirements, testing, troubleshooting, and best-practice guides** — including a failure-mode matrix

covering the unzip error, the missing sign-in tile, reboot-loop and IME-state loss, ESP hangs, the Win10-vs-Win11 mismatch, and hybrid first-logon failures.

Business outcomes

- **Hybrid Autopilot works over a user-initiated VPN** — the tunnel is up before logon, so domain join, Group Policy, and first-time logon succeed off-network, with no Always On dependency.
- **The fragile seams are removed** — one correctly sequenced app, ESP-owned reboot, and an AV exclusion

end the intermittent unzip failures and the missing Connect Before Logon tile.

- **Faster diagnosis** — a repeatable log-capture runbook replaces guesswork in the test window.
- **Lower risk** — piloted on Windows 11 before fleet-wide rollout, validated against explicit gates, and reversible at any point.

Scope and boundary

The solution is authored at full detail on the **Microsoft Intune, Autopilot, Entra ID, and Windows** side. The **GlobalProtect portal / gateway configuration and the GlobalProtect app internals** are the customer's existing Palo Alto environment and follow **Palo Alto Networks' official documentation** — this package consumes those artifacts (the MSI, portal FQDN, and authentication method) and delivers Connect Before Logon reliably during enrollment.

Bottom line

GlobalProtect Connect Before Logon, installed and registered during Hybrid-Join Autopilot, with the sign-in tile present at first logon — delivered as a single sequenced app that lets ESP own the reboot, made diagnosable with a log-capture runbook, and proven on a Windows 11 pilot before you scale.

Solution Overview — GlobalProtect Connect Before Logon (PLAP) for Hybrid Azure AD Join Autopilot via Microsoft Intune

What this solution is

This solution deploys the **Palo Alto Networks GlobalProtect** app with its **Connect Before Logon (CBL)** capability to your **Windows 11** fleet using **Microsoft Intune**, so the VPN tunnel comes up **at the Windows sign-in screen — before the user signs in to Windows.** It is built for the specific, hard case: a **Hybrid Azure AD Join Windows Autopilot** provisioning where the device must reach a **domain controller before first user logon**, but the organization has moved to **user-initiated VPN** and can no longer rely on Always On VPN. The goal is a consistent, tested, hands-off build: the GlobalProtect app, the Connect Before Logon registration, and the portal configuration all land during Autopilot enrollment, are validated on a small pilot first, and are then expanded fleet-wide with a documented rollback path — and, critically, are **diagnosable** when the enrollment sequence misbehaves.

The work is authored at click-by-click detail on the **Intune, Autopilot, Entra ID, and Windows** side. The **GlobalProtect portal / gateway configuration and the GlobalProtect app internals are deferred to Palo Alto Networks' official documentation** — this solution tells you *what* Palo Alto artifacts you need (the MSI, the portal FQDN, the authentication method) and *where* they plug into Intune, not how to click

through Palo Alto's own consoles.

The real-world scenario this solves

The situation this package is built around is a live deployment blocker, not a hypothetical:

• The organization **abandoned Always On VPN after a security incident** and now requires

user-initiated VPN — a tunnel the user (or the sign-in screen) brings up, not one that is always established by the machine.

• The organization provisions Windows with **Hybrid Azure AD Join Windows Autopilot (user-driven)**.

Hybrid join and first interactive logon need **line-of-sight to a domain controller before the user signs in** — for domain join completion, Group Policy, and a first-time (no cached credential) logon.

• User-initiated VPN breaks that pre-logon reachability **unless** the tunnel can be raised *at the

sign-in screen*. **GlobalProtect Connect Before Logon (PLAP)** is exactly that capability — it adds a network sign-in tile to the Windows lock screen so the tunnel comes up before Windows logon.

• In practice the base GlobalProtect client deploys fine from Intune, but the **PLAP registration** step

is fragile: it **intermittently fails with an "unzipping" error** during Autopilot, reboot orchestration is suspected of compounding the failure, and the **"Connect Before Logon" control is not reliably visible** at the sign-in screen. Early testing was on **Windows 10**, while production targets **Windows 11**.

This solution makes that scenario **deployable** (a single, correctly sequenced Intune app that installs, enables, and registers Connect Before Logon and lets the Enrollment Status Page own the reboot) **and diagnosable** (a log-capture runbook and validation tooling for a Windows 11 test device), so the "unzip" failure, the reboot fragility, and the missing sign-in tile are each turned into a known cause with a known fix.

What GlobalProtect Connect Before Logon (CBL / PLAP) is

Connect Before Logon (CBL) is implemented on Windows as a **Pre-Login Access Provider (PLAP)** — a *connectable credential provider* that adds a **network sign-in / Connect Before Logon** tile to the Windows sign-in screen, so a user can establish the GlobalProtect tunnel **before** authenticating to Windows. It was introduced in the GlobalProtect app **5.2**, so this solution requires **GlobalProtect app 5.2 or later** and targets **Windows 11**. CBL supports basic credentials, certificates, and SAML authentication (the head-end decides which — deferred to Palo Alto docs).

For that tile to appear and work, **three things** must be true on the device, in order:

1. **The GlobalProtect app (5.2+) is installed.**
2. **Connect Before Logon is enabled and pointed at a portal.** This is a registry setting under

`HKEY_LOCAL_MACHINE\SOFTWARE\Palo Alto Networks\GlobalProtect\PanSetup\` — the value `Prelogon` set to `1`, and the default `Portal` address set under the same `PanSetup` key. Both must exist **before** registration and before the sign-in screen is reached.

1. **The PLAP credential provider is registered.** Registration is performed by running

`PanGPS.exe -registerplap` (from `C:\Program Files\Palo Alto Networks\GlobalProtect`) in the **SYSTEM** context. (The reverse is `PanGPS.exe -unregisterplap`.) The GlobalProtect service (**PanGPS**) must be ready before registration runs.

Miss any one of the three — app below 5.2, `Prelogon` not set, or `-registerplap` never run (or run before the service is ready) — and the Connect Before Logon tile silently never appears. This solution delivers **all three together** so a device is either fully Connect-Before-Logon-ready or clearly not — never half-configured.

How it is delivered through Intune (the recommended single Win32 app)

Everything is delivered from the cloud to Autopilot-provisioning Windows 11 devices. Palo Alto ships a wrapper-script pattern (`InstallGlobalProtect_PLAP.ps1`) that installs GlobalProtect, sets the portal, and registers PLAP; this solution **recommends consolidating install + registry + registration into ONE Win32 app** so the ordering and reboot cannot fragment across separate deployments:

• **The single Win32 app.** One **Win32 app** (`.intunewin`) that:

1. installs the **GlobalProtect MSI (5.2+)** with `Start-Process … -Wait` so the installer fully completes before the script returns,

1. writes the `PanSetup` **registry** values — `Prelogon = 1` and the default `Portal` address,

2. **waits for the GlobalProtect service (PanGPS) to be ready**, then runs `PanGPS.exe -registerplap` in the **SYSTEM** context, and

1. **returns exit code** `3010` (soft reboot) so **Intune / the Enrollment Status Page manages the reboot** — the script never calls a forced `Restart-Computer` itself.

• **App name:** `GlobalProtect – Connect Before Logon (PLAP) – Win32`

• **The Enrollment Status Page (ESP) blocking app.** The Win32 app is made a **blocking app** in the Autopilot ESP so provisioning **waits** for Connect Before Logon to be installed and registered before the user reaches the desktop.

• **ESP reference:** `ESP – Autopilot – GlobalProtect PLAP Blocking`

• **(Optional) the device certificate** (only if the portal uses certificate authentication). A SCEP certificate profile issuing a device certificate the portal trusts.

• **Certificate profile:** `GlobalProtect – PLAP – Device Cert (SCEP) – Windows`

The two-app alternative (documented, not recommended)

You *can* split the work into a base app plus a registration app — `GlobalProtect – VPN Client – Win32` (installs GlobalProtect + sets the `PanSetup` registry) and `GlobalProtect – PLAP Registration – Win32` (runs `PanGPS.exe -registerplap`), with the registration app declared **dependent** on the base app. This works, but it introduces an ordering/reboot seam between two deployments during Autopilot — the exact seam where the "unzip" and sequencing failures cluster. Prefer the **single-app** path; the two-app model is documented for environments that already separate them.

Why exit code 3010 and not a scripted reboot (the reboot-sequencing rule)

The most damaging mistake in this scenario is calling a **forced** `Restart-Computer` inside the install script during the Autopilot ESP. That interrupts the Intune Management Extension (IME) agent mid-state and causes install failures or lost app state — which is what makes reboot orchestration *look* like the

cause of the "unzip" error. The correct pattern is:

• **Never** reboot from inside the script. **Return** `3010` (soft reboot) and let **Intune/ESP** own the reboot at the right moment in the sequence.

• **Map the return codes** in the Win32 app: `0` and `1707` = success, `3010` = soft reboot, `1618` = retry (another install in progress).

• **Use** `Start-Process … -Wait` for the MSI so the app never reports complete before the installer actually finishes (a missing `-Wait` makes the app "succeed" early, before PLAP can register).

The "unzipping" error, in one paragraph

The intermittent **"unzipping"** failure is Intune Win32 error `0x87D30067`: the IME downloaded the `.intunewin` but **failed to unzip it** into the IME content/cache path (`C:\Program Files (x86)\Microsoft Intune Management Extension\Content\…` / IMECache). The usual causes are **antivirus (Microsoft Defender or third-party) locking or scanning the IMECache during extraction**, **low disk space**, a **corrupt or improperly re-wrapped** `.intunewin` (for example nesting an already-wrapped payload), or path/permission tampering. The fixes: **exclude the IME content/IMECache paths from AV during provisioning**, free disk space, **re-wrap the package cleanly** with the current **Microsoft Win32 Content Prep Tool**, and verify the source folder. The Troubleshooting guide turns each cause into a concrete step.

Autopilot enrollment-time targeting (why an Assigned device group with an SP owner)

Because the app must install **during** Autopilot (before a user is on the device), it is targeted at an Entra **device** group that is populated **at enrollment**. Enrollment-time device grouping requires an **Assigned** (not Dynamic) security group whose **Owner** is the **Intune Provisioning Client** first-party service principal (**AppId** `f1346770-5b25-470b-88bd-d5744ab7952c`, shown in many tenants as *Intune Autopilot ConfidentialClient*). Without that service principal as an owner, the assigned group stays **empty at enrollment** and nothing deploys. This solution names those groups and sets the owner explicitly:

• **Pilot device group:** `PILOT – GlobalProtect PLAP – Autopilot Devices`

• **Production device group:** `PROD – GlobalProtect PLAP – Autopilot Devices`

Benefits

• **The tunnel is up before Windows logon.** Hybrid Autopilot devices reach a domain controller for domain join, Group Policy, and first-time (no cached credential) logon — over a **user-initiated** VPN, with no Always On dependency.

• **One correctly sequenced app.** Install, registry, service-readiness wait, and PLAP registration in a single Win32 app that returns `3010` and lets ESP own the reboot — the fragile seams are removed.

• **Diagnosable, not just deployable.** A log-capture runbook and validation tooling for a Windows 11 test device turn the "unzip" error, the reboot fragility, and the missing sign-in tile into known causes with known fixes.

• **Pilot-first and rollback-ready.** App, ESP blocking, and any certificate profile are assigned to a named pilot device group, validated against explicit gates, and only then expanded. Uninstall/exclude is a one-click reversal.

• **Traceable and reportable.** Every object is named to a convention, so the discovery, validation, log-capture, and reporting scripts find and check them deterministically — including whether the Connect Before Logon registration is actually present.

Limitations and boundaries

• **Windows only, Windows 11 target.** Connect Before Logon is a Windows credential-provider (PLAP) feature; this solution targets **Windows 11** (production) with GlobalProtect **5.2+**. Windows 10 may work but early "works on Win10, not Win11" reports mean you must **re-test on Windows 11** — see Troubleshooting.

• **GlobalProtect portal / gateway and app internals are out of scope.** The GlobalProtect **portal and gateway **configuration, the authentication method the portal enforces, and the GlobalProtect** app **internals are done per** Palo Alto Networks' official documentation**. This solution consumes those artifacts (the MSI, the portal FQDN, the auth method) and delivers them via Intune.

• **Autopilot Hybrid Azure AD Join is a prerequisite, not a deliverable.** The Autopilot profile, the Enrollment Status Page, and the Domain Join configuration profile / on-prem Domain Join (ODJ) connector must already be configured (Requirements guide). Hybrid join also needs the device to reach a DC — which is exactly what Connect Before Logon provides at the sign-in screen.

• **A working PKI is assumed only if the portal uses certificate auth.** The optional device certificate depends on a SCEP path (on-prem NDES, Intune Cloud PKI, or a supported third-party CA connector).

The standard deployment model

This solution follows the category-standard model, with explicit pass/fail gates at each step:

Assess → **Configure** → **Pilot** → **Validate** → **Expand**

1. **Assess.** Confirm prerequisites (Requirements guide), gather the Palo Alto inputs (the GlobalProtect 5.2+ MSI, the portal FQDN, the auth method), and confirm the Autopilot Hybrid-Join profile, the ESP, and the Domain Join config / ODJ connector are in place. **Gate A.**

1. **Configure.** Build the single Win32 app (install + `PanSetup` registry + service-readiness wait + `PanGPS.exe -registerplap` in SYSTEM + return `3010`), set install context System, detection and return-code mapping, add it as an **ESP blocking app**, and assign to `PILOT – GlobalProtect PLAP – Autopilot Devices`; add the AV/IMECache exclusion. **Gate B.**

1. **Pilot.** Run an Autopilot Hybrid-Join provisioning on a **Windows 11** test device. **Gate C.**

2. **Validate.** Confirm the Connect Before Logon tile appears at the sign-in screen, the pre-logon tunnel comes up, the DC is reachable, and a first interactive **hybrid domain** logon succeeds off-network; collect logs with the toolkit. **Gate D.**

1. **Expand.** Widen assignment to `PROD – GlobalProtect PLAP – Autopilot Devices`, monitor coverage, and

keep the rollback ready. **Gate E.**

Proceed to the **Requirements guide** (`02-requirements.md`) to confirm you are ready to build.

Requirements — GlobalProtect Connect Before Logon (PLAP) for Hybrid Azure AD Join Autopilot via Microsoft Intune

Confirm **every** requirement below before you build. Each entry follows the same shape: **Description / Why required / How to verify / How to implement if missing.** The category-level prerequisites (Entra tenant, Intune, admin roles, pilot group, change control) are in `global-requirements-network.md`; the items here are specific to deploying **GlobalProtect Connect Before Logon (PLAP)** during a **Hybrid Azure AD Join Windows Autopilot** provisioning.

> ***Vendor boundary:*** *the **GlobalProtect portal / gateway configuration and the GlobalProtect app internals are deferred to Palo Alto Networks' official documentation.** Below, "obtain from Palo Alto" means *get the artifact/value from your Palo Alto admin or Palo Alto docs* — this guide does not fabricate GlobalProtect portal menu paths.*

> ***Government note (GCC High / DoD):*** *the steps below name the commercial portals (`https://intune.microsoft.com`, `https://entra.microsoft.com`, `https://login.microsoftonline.com`). In a US-Gov (GCC High / DoD) tenant use the sovereign portals instead — `https://intune.microsoft.us` / `https://endpoint.microsoft.us`, `https://entra.microsoft.us`, and `https://login.microsoftonline.us` — and **verify each step for GCC High**, as feature availability and rollout can differ. This guide does not invent GCC-specific menus.*

R1. Windows 11 scope confirmed

Description: The target devices are **Windows 11** devices provisioned by Windows Autopilot; Connect Before Logon is a Windows-only feature and production here is Windows 11. **Why required:** Connect Before Logon is implemented as a Windows credential provider (PLAP). Early testing on Windows 10 can behave differently from Windows 11 (see Troubleshooting), so the pilot and the log-capture runbook must run on **Windows 11**. **How to verify:** Your in-scope device list is Windows 11; the pilot test device is Windows 11. **How to implement if missing:** Scope the pilot and production device groups to Windows 11. If a Windows 10 estate remains, re-test the whole build separately on Windows 10 rather than assuming parity.

R2. Microsoft Intune licensing and Autopilot-enrolled devices

Description: An Intune-bearing license plan (e.g. Microsoft 365 E3/E5, EMS, or Intune standalone) and the target Windows 11 devices registered for Windows Autopilot. **Why required:** Intune is the delivery plane for the Win32 app, the Enrollment Status Page, and any certificate profile. Nothing deploys to a device that is not enrolled and licensed. **How to verify:** In the **Microsoft Intune admin center** (`https://intune.microsoft.com`), go to **Devices > Enrollment > Windows > Windows Autopilot devices** and confirm the target devices are registered; under **Tenant administration > Licenses**, confirm

Intune licenses are assigned. **How to implement if missing:** Assign an Intune-bearing license; register the devices for Autopilot (hardware hash import or OEM/CSP registration).

R3. Windows Autopilot Hybrid Azure AD Join deployment profile

Description: A **Windows Autopilot deployment profile** configured for **Hybrid Azure AD Join** (user-driven), assigned to the Autopilot device group. **Why required:** This solution exists to make hybrid-join Autopilot work over a user-initiated VPN. Without the hybrid-join Autopilot profile there is no provisioning sequence for Connect Before Logon to support. **How to verify:** Intune → **Devices > Enrollment > Windows > Deployment Profiles**; a profile exists with **Join type = Hybrid Azure AD joined**, assigned to the Autopilot device group. **How to implement if missing:** Create the deployment profile (Intune → **Devices > Enrollment > Windows > Deployment Profiles > Create profile > Windows PC**), set **Join to Azure AD as = Hybrid Azure AD joined**, and assign it to the Autopilot device group.

R4. Domain Join configuration profile and on-prem Domain Join (ODJ) connector

Description: An Intune **Domain Join** device-configuration profile (target domain, OU) and a healthy **Intune Connector for Active Directory** (the on-prem Domain Join / ODJ connector). **Why required:** Hybrid Azure AD Join Autopilot completes the on-prem domain join through the ODJ connector; the device must reach a domain controller to finish join and first logon — the reachability that Connect Before Logon provides at the sign-in screen. **How to verify:** Intune → **Devices > Configuration**; a **Domain Join** profile exists and is assigned. Intune → **Devices > Enrollment > Windows > Intune Connector for Active Directory**; the connector shows **Active**. **How to implement if missing:** Install/repair the Intune Connector for Active Directory on a domain-joined server, create the **Domain Join** profile (domain, OU), and assign it to the Autopilot device group.

R5. Enrollment Status Page (ESP) configured

Description: A Windows **Enrollment Status Page** profile assigned to the Autopilot devices, into which the GlobalProtect PLAP app will be added as a **blocking app**. **Why required:** The ESP is what makes provisioning **wait** for Connect Before Logon to install and register before the user reaches the desktop, and it is what owns the reboot the app requests with exit code `3010`. **How to verify:** Intune → **Devices > Enrollment > Windows > Enrollment Status Page**; a profile (other than the disabled default, if you use a targeted one) is enabled and assigned. **How to implement if missing:** Create/enable an ESP profile (Intune → **Devices > Enrollment > Windows

*Enrollment Status Page > Create**), set** Show app and profile installation progress = Yes**, and*

assign it. You will reference it as `ESP – Autopilot – GlobalProtect PLAP Blocking` in Implementation.

R6. GlobalProtect app installer (5.2 or later) — the MSI

Description: The official **GlobalProtect** Windows **MSI** installer, version **5.2 or later** (the first release with Connect Before Logon). **Why required:** Connect Before Logon (the PLAP credential provider) exists only in GlobalProtect 5.2+. The MSI is the exact binary the Win32 app installs. **How to verify:** Obtain the MSI

from your Palo Alto admin / Palo Alto support portal and confirm the version is ≥ **5.2** for the release your portal supports. **How to implement if missing:** Obtain the correct entitled GlobalProtect MSI (5.2+) from Palo Alto. Do not repackage or re-sign the vendor binary; use the vendor-provided MSI.

R7. GlobalProtect portal FQDN and authentication method

Description: The **portal FQDN** (e.g. `portal.your-company.com`) that Connect Before Logon points at, and the **authentication method** the portal enforces (basic credentials, certificate, or SAML). **Why required:** The portal FQDN is written into the `PanSetup` registry (`Portal`) and the auth method determines whether an optional device certificate (R8) is needed. These are inputs, not things this guide configures on the Palo Alto side. **How to verify:** Obtain from your Palo Alto admin: the **portal FQDN** and the **pre-logon auth method**. Confirm a manual GlobalProtect connect to that portal already succeeds with the intended method. **How to implement if missing:** Configure the GlobalProtect **portal/gateway** and its authentication **per Palo Alto Networks' official documentation** — Palo Alto-side work outside this solution's scope. Record the portal FQDN and auth method for Implementation.

R8. Certificate infrastructure (only if the portal uses certificate authentication)

Description: A working PKI path for Intune to deliver a **device** certificate — an on-prem **NDES/SCEP** connector, **Intune Cloud PKI**, or a supported third-party CA connector — plus the trusted root/intermediate CA. **Why required:** If the portal authenticates Connect Before Logon with a certificate, each device needs a machine certificate the portal trusts, delivered by Intune, present before the sign-in screen. (Skip R8 if the portal uses basic credentials or SAML at the Connect Before Logon window.) **How to verify:** Intune → **Devices > Configuration**; a SCEP certificate profile named `GlobalProtect – PLAP – Device Cert (SCEP) – Windows` can be created and deploys to a test device, and the issued cert chains to a CA the GlobalProtect portal trusts. **How to implement if missing:** Stand up the SCEP path (NDES + Intune Certificate Connector, or enable Intune Cloud PKI), publish the trusted root/intermediate, then create the SCEP profile. Ensure the **issuing CA is trusted by the GlobalProtect portal/gateway** (Palo Alto-side trust config is deferred to Palo Alto docs).

R9. Autopilot device group — Assigned, owned by the Intune Provisioning Client service principal

Description: A pilot Autopilot **device** group `PILOT – GlobalProtect PLAP – Autopilot Devices` and a production device group `PROD – GlobalProtect PLAP – Autopilot Devices`, each an **Assigned** (not Dynamic) security group whose **Owner** includes the **Intune Provisioning Client** service principal (**AppId** `f1346770-5b25-470b-88bd-d5744ab7952c`). **Why required:** For a device to receive the app **at enrollment**, it must be in the target group at enrollment time. Enrollment-time device grouping works only when the group is **Assigned** and the Intune Provisioning Client SP **owns** the group; without that owner the assigned group stays empty at enrollment and the app never deploys. **How to verify:** Entra → **Groups > All groups**; `PILOT – GlobalProtect PLAP – Autopilot Devices` exists, is **Assigned**, and under **Owners** lists the service principal with AppId `f1346770-5b25-470b-88bd-d5744ab7952c` (search by **AppId** — the display name, often *Intune Autopilot ConfidentialClient*, varies by tenant). **How to implement if missing:** Entra → **Groups > New group > Security** (membership type **Assigned**); name it exactly `PILOT – GlobalProtect PLAP – Autopilot Devices`; add the Autopilot devices; then **Owners > Add owners** and add the SP by **AppId**

`f1346770-5b25-470b-88bd-d5744ab7952c`. If the SP is absent, provision it by opening the Intune portal once as Global Administrator. The included `GPPLAP-Set-ScopeTargeting.ps1` creates the group, adds devices, and sets the SP owner for you.

R10. Reboot-sequencing and return-code design confirmed

Description: A confirmed design that the install script **never** calls a forced `Restart-Computer` during the ESP, **returns exit code** `3010` for reboot, uses `Start-Process … -Wait` for the MSI, and maps the Win32 return codes (`0`/`1707` success, `3010` soft reboot, `1618` retry). **Why required:** A scripted reboot mid-ESP interrupts the Intune Management Extension and causes install failures/lost state — the failure mode this solution is built to remove. Returning `3010` lets Intune/ESP own the reboot; `-Wait` stops the app reporting complete before the installer finishes. **How to verify:** Your install script has no `Restart-Computer`; it returns `3010` after registration; the MSI is launched with `-Wait`; the app's return-code mapping is set (Implementation §2). **How to implement if missing:** Adopt the single-app pattern in Implementation §2 (it already encodes this). Never reboot from the script; always return `3010` and map the codes on the app.

R11. Antivirus exclusion for the IME content/cache path (to avoid the "unzip" 0x87D30067 failure)

Description: During provisioning, the Intune Management Extension content/cache path is **excluded** from real-time antivirus scanning (Microsoft Defender or third-party), and the device has adequate free disk space. **Why required:** The intermittent **"unzipping"** failure is Intune error `0x87D30067` — the IME failed to unzip the `.intunewin` into its content/cache path, most often because AV locked or scanned the IMECache during extraction (or the disk was low, or the package was badly re-wrapped). **How to verify:** Your provisioning AV policy excludes `C:\Program Files (x86)\Microsoft Intune Management Extension\Content` (and the IMECache), and the target devices have sufficient free disk. **How to implement if missing:** Add the IME content/IMECache path to the AV exclusion list applied during provisioning; ensure free disk space; and always re-wrap the `.intunewin` cleanly with the current Content Prep Tool (R12) rather than nesting an already-wrapped payload.

R12. Microsoft Win32 Content Prep Tool (current version)

Description: The **Microsoft Win32 Content Prep Tool** (`IntuneWinAppUtil.exe`), current release, on the packaging workstation, used to wrap the GlobalProtect MSI + the install/registration script into an `.intunewin`. **Why required:** A single Win32 app that installs the MSI, writes the registry, waits for the service, and registers PLAP is best delivered as an `.intunewin`. A stale tool or a nested/already-wrapped payload is a known cause of the `0x87D30067` unzip failure. **How to verify:** `IntuneWinAppUtil.exe` runs and reports its version; you wrap a **clean** source folder (the MSI + the script), not an already-wrapped file. **How to implement if missing:** Download the current Win32 Content Prep Tool from Microsoft's official `microsoft-Win32-Content-Prep-Tool` release and place it on the packaging workstation.

R13. Administrative roles for building the deployment

Description: Intune admin rights to create Win32 apps, the ESP, and certificate profiles; Entra rights to create/populate groups and set group owners (Groups Administrator or better). **Why required:** Packaging

the app, editing the ESP, creating cert profiles, and setting the service-principal group owner all require elevated roles. **How to verify:** In Intune you can reach **Apps > Add** and **Devices > Enrollment**; in Entra you can create groups and add owners. **How to implement if missing:** Have a Global Administrator assign **Intune Administrator** and **Groups Administrator** (least-privilege) to the operator; setting the first-party SP as a group owner may require a Global Administrator the first time.

R14. Microsoft Graph PowerShell SDK (to run the included scripts)

Description: The **Microsoft.Graph** PowerShell SDK installed on the operator workstation, used by the discovery/report/scope-targeting scripts (the on-device validation and log-capture scripts run locally and need no Graph). **Why required:** The scripts call Microsoft Graph (Intune/Entra) to inventory the app, the ESP, the cert profile, the device groups, group membership, and the group owner. **How to verify:** In PowerShell, `Get-Module Microsoft.Graph -ListAvailable` returns a version. **How to implement if missing:** Run `Install-Module Microsoft.Graph -Scope CurrentUser`. The scripts call `Connect-MgGraph` with the specific delegated scopes each one documents in its header.

R15. Change control and rollback (category requirement, restated)

Description: An approved change window and a written rollback (exclude the group / remove the app from the ESP blocking list / uninstall the app / unregister PLAP). **Why required:** Adding a blocking app to the Autopilot ESP and a credential provider to the sign-in screen changes the provisioning and logon experience for real users; changes must be reversible. **How to verify:** A rollback procedure exists and is approved; you know how to exclude `PILOT – GlobalProtect PLAP – Autopilot Devices`, remove the app from the ESP blocking list, and flip the Win32 app to **Uninstall** (which can also run `PanGPS.exe -unregisterplap`). **How to implement if missing:** Document the rollback (assignment exclusion, remove from ESP blocking apps, app **Uninstall** intent incl. `-unregisterplap`, cert removal) and the comms plan before you assign anything.

When all requirements pass, proceed to `03-implementation.md`.

Implementation — GlobalProtect Connect Before Logon (PLAP) for Hybrid Azure AD Join Autopilot via Microsoft Intune

This is the ordered, click-by-click build. Follow the phases in order: **Assess** → **Configure** → **Pilot** → **Validate** → **Expand.** Do **not** skip the pilot. Use the **exact object names** shown — later steps, tests, troubleshooting, and scripts reference them verbatim.

> ***Vendor boundary:*** *the GlobalProtect* ***portal / gateway*** *configuration, the authentication method the portal enforces, and the GlobalProtect* ***app*** *internals are* ***deferred to Palo Alto Networks' official documentation****. This guide covers the Intune, Autopilot, Entra, and Windows side at full detail and *consumes* the Palo Alto-supplied* ***MSI (5.2+)****,* ***portal FQDN****, and* ***auth method*** *as inputs.*
>
> ***Government (GCC High / DoD):*** *substitute the sovereign portals —* `https://intune.microsoft.us` / `https://endpoint.microsoft.us`*,* `https://entra.microsoft.us`*,* `https://login.microsoftonline.us` *— and* ***verify each***

***step for GCC High**. Menu names below are the commercial names.*

Object naming convention (use verbatim)

Object	Name
Pilot device group (Assigned; SP owner)	`PILOT – GlobalProtect PLAP – Autopilot Devices`
Production device group (Assigned; SP owner)	`PROD – GlobalProtect PLAP – Autopilot Devices`
Win32 app (install + enable + register — recommended)	`GlobalProtect – Connect Before Logon (PLAP) – Win32`
Alt: base app (two-app model)	`GlobalProtect – VPN Client – Win32`
Alt: registration app (two-app model)	`GlobalProtect – PLAP Registration – Win32`
Optional device certificate profile (if cert auth)	`GlobalProtect – PLAP – Device Cert (SCEP) – Windows`
ESP blocking-app reference	`ESP – Autopilot – GlobalProtect PLAP Blocking`
Assignment filter (optional)	`Filter – GlobalProtect PLAP – Windows`

Phase 1 — Assess

1. Confirm every item in `02-requirements.md` passes.
2. Gather the Palo Alto inputs (from your Palo Alto admin, per Palo Alto docs):

• The **GlobalProtect MSI**, version **5.2 or later** (R6).

• The **portal FQDN** (e.g. `portal.your-company.com`) (R7).

• The **authentication method** the portal enforces for Connect Before Logon (basic / certificate / SAML) (R7) — this decides whether the optional certificate profile (R8) is needed.

1. Confirm the Autopilot side is ready: the **Hybrid Azure AD Join** deployment profile (R3), the **Domain Join** profile + **ODJ connector** *Active* (R4), and the **Enrollment Status Page** (R5).
1. Confirm the **reboot-sequencing design** (R10) and the **AV/IMECache exclusion** (R11) are in place.
2. Run the discovery script to baseline readiness:

`powershell -ExecutionPolicy Bypass -File .\scripts\GPPLAP-Discovery.ps1` Review its readiness table before building anything.

Gate A (Assess): All requirements pass; the GlobalProtect 5.2+ MSI, portal FQDN, and auth method are recorded; the Autopilot Hybrid-Join profile, Domain Join + ODJ connector, and ESP are confirmed; and discovery shows the tenant is ready. If not, stop and remediate.

Phase 2 — Configure

2.0 Create the Autopilot device group(s) — Assigned, with the Provisioning Client SP as owner

Use the script (recommended) or the portal. The group **must be Assigned** and **owned by the Intune Provisioning Client service principal** so it populates at enrollment (R9).

- **Script:** `powershell -ExecutionPolicy Bypass -File .\scripts\GPPLAP-Set-ScopeTargeting.ps1`

(creates `PILOT – GlobalProtect PLAP – Autopilot Devices` and, with `-CreateProd`, `PROD – GlobalProtect PLAP – Autopilot Devices` as **Assigned** groups, adds the devices you supply by name, and sets the SP `f1346770-5b25-470b-88bd-d5744ab7952c` as an **owner**). Defaults to `-WhatIf` preview; re-run with `-WhatIf:$false` to apply.

- **Portal:**

1. Entra admin center → **Groups > New group** → **Group type: Security**, **Membership type:

Assigned → **name it exactly** `PILOT – GlobalProtect PLAP – Autopilot Devices` → **add the Autopilot** devices **as members** → Create**.

1. Open the group → **Owners > Add owners** → search by **AppId**

`f1346770-5b25-470b-88bd-d5744ab7952c` (the display name, often *Intune Autopilot ConfidentialClient*, varies) → **Select**. If the SP is not found, open the Intune portal once as a **Global Administrator** to provision it, then retry.

1. Repeat for `PROD – GlobalProtect PLAP – Autopilot Devices` (production is assigned in Phase 5).

2.1 (Optional) Create the assignment filter

If a group spans device types, create a Windows filter to keep this build on Windows only.

1. Intune → **Tenant administration > Filters > Create**.
2. Name it exactly `Filter – GlobalProtect PLAP – Windows`; **Platform: Windows 10 and later**.
3. Set a rule (e.g. `(device.osVersion -startsWith "10.0.2")` to bias toward Windows 11, or `(device.operatingSystem -eq "Windows")`); **Review + create.** Attach it to the app assignment where you want to scope by OS.

2.2 (If cert auth) Create the device certificate profile

Only if R8 applies (the portal authenticates Connect Before Logon with a certificate).

1. First deploy the **Trusted Root CA** so the device trusts the issuing CA: Intune → **Devices >

Configuration > Create > New Policy → Windows 10 and later → Templates > Trusted certificate** → upload your root/intermediate `.cer`, set the destination store, and assign to the pilot device group.

1. Intune → **Devices > Configuration > Create > New Policy** → **Windows 10 and later** → **Templates >

SCEP certificate **(or** PKCS certificate**).

1. Name it exactly `GlobalProtect – PLAP – Device Cert (SCEP) – Windows`.

2. Set **Certificate type: Device** (there is no user context at the sign-in screen), **Subject name

format**,** SAN**,** Key usage**,** Extended key usage = Client Authentication **(**`1.3.6.1.5.5.7.3.2`**), a** renewal threshold **comfortably before expiry, the** SCEP server URL(s)**, and the** root CA** from step 1.

1. **Assign** to `PILOT – GlobalProtect PLAP – Autopilot Devices`. (The issuing CA must be trusted by the

GlobalProtect portal/gateway — Palo Alto-side trust config is deferred to Palo Alto docs.)

2.3 Build the single Win32 app — install + PanSetup registry + service-readiness wait + register PLAP

This is the recommended path: **one** app that does everything, so ordering and reboot cannot fragment across separate deployments.

2.3a Prepare the package (install + registry + registration in one script, returning 3010)

1. Place into a **clean** source folder (not an already-wrapped file):

• the **GlobalProtect MSI (5.2+)** from R6, and

• an **install script** (`Install-GlobalProtect-PLAP.ps1`) that consolidates the steps Palo Alto's

`InstallGlobalProtect_PLAP.ps1` wrapper performs.

1. The install script must, **in order**:

2. Install the MSI with a **blocking** call so it fully completes before the script returns —

`Start-Process msiexec.exe -ArgumentList '/i "GlobalProtect64.msi" /qn' -Wait`. **Never** omit `-Wait` (the app would report complete before PLAP can register).

1. Write the `PanSetup` **registry** values that enable Connect Before Logon and set the portal —

under `HKLM:\SOFTWARE\Palo Alto Networks\GlobalProtect\PanSetup`:

• `Prelogon` = `1` (DWORD)

• `Portal` = the portal FQDN from R7 (String)

1. **Wait for the GlobalProtect service (**`PanGPS`**) to be ready** — poll `Get-Service PanGPS` until it

is *Running* (with a bounded timeout) so registration does not run before the service exists.

1. Run the PLAP registration **in the SYSTEM context** (the Win32 app already runs as System, §2.3c):

```
Start-Process "C:\Program Files\Palo Alto Networks\GlobalProtect\PanGPS.exe"
-ArgumentList '-registerplap' -Wait
```

1. `exit 3010` to request a **soft reboot** that **Intune/ESP** manages. **Never** call

`Restart-Computer` in the script.

1. Wrap the **clean source folder** into an `.intunewin` with the **current** Microsoft Win32 Content

Prep Tool (R12): `IntuneWinAppUtil.exe -c your-source-folder -s Install-GlobalProtect-PLAP.ps1 -o your-output-folder` Re-wrap cleanly every time — do **not** nest an already-wrapped payload (a known cause of the unzip `0x87D30067` failure).

2.3b Add the app to Intune

1. Intune admin center → **Apps > Windows > Add** → App type **Windows app (Win32)** → upload the `.intunewin`.

1. **App information.** Name it exactly `GlobalProtect – Connect Before Logon (PLAP) – Win32`; set

publisher = Palo Alto Networks, version (the MSI version), and a description noting it installs GlobalProtect, enables Connect Before Logon (`Prelogon=1` + portal), and registers PLAP.

2.3c Program, context, requirements, detection, and return codes

1. **Program.**

• **Install command:**

```
powershell.exe -ExecutionPolicy Bypass -File Install-GlobalProtect-PLAP.ps1
```

• **Uninstall command:** MSI product-code uninstall (optionally preceded by `PanGPS.exe -unregisterplap`).

• **Install behavior: System.** (PLAP registration must run as **SYSTEM**.)

• **Device restart behavior:** *Determine behavior based on return codes* (so the `3010` mapping in the next step drives the reboot).

1. **Return codes (map exactly):** `0` = Success, `1707` = Success, `3010` = **Soft reboot**, `1618` = **Retry**. This is what lets ESP own the reboot instead of the script.

1. **Requirements.** OS architecture **x64**; minimum OS version per Palo Alto's support matrix (target Windows 11).

1. **Detection rules.** Detect that the app installed **and** Connect Before Logon is enabled and registered — combine:

• the GlobalProtect install (MSI product code, or file/version of a GlobalProtect binary confirming ≥ **5.2**),

• the registry value `HKLM:\SOFTWARE\Palo Alto Networks\GlobalProtect\PanSetup\Prelogon` = `1`, and

• the PLAP credential-provider registration (a Cisco/Palo-Alto-style provider key exists under `HKLM:\SOFTWARE\Microsoft\Windows\CurrentVersion\Authentication\Credential Providers`). Detecting only the MSI marks a device *Installed* even when Connect Before Logon is not registered.

1. **Assignments.** Under **Required**, add `PILOT – GlobalProtect PLAP – Autopilot Devices` (optionally

with `Filter – GlobalProtect PLAP – Windows`). Leave production for Phase 5. **Review + create.**

2.3d (Alternative) Two-app model — documented, not recommended

If your environment already separates install from registration, build two apps instead of one:

- `GlobalProtect – VPN Client – Win32` — installs the MSI and writes the `PanSetup` registry

(`Prelogon=1` + `Portal`), returning `3010`.

- `GlobalProtect – PLAP Registration – Win32` — waits for `PanGPS` readiness and runs

`PanGPS.exe -registerplap` in SYSTEM, returning `3010`. Under this app's **Dependencies**, add `GlobalProtect – VPN Client – Win32` as a required dependency so registration cannot run before the client is installed.

Prefer the single app (§2.3a–c): splitting the work reintroduces the ordering/reboot seam where the unzip and sequencing failures cluster.

2.4 Add the app as an ESP blocking app

1. Intune → **Devices > Enrollment > Windows > Enrollment Status Page** → open your ESP profile (you

will reference it as `ESP – Autopilot – GlobalProtect PLAP Blocking`).

1. Set **Block device use until required apps are installed = Yes**.
2. Under **Select the apps that must be installed before the device can be used**, add

`GlobalProtect – Connect Before Logon (PLAP) – Win32` (and, if you used the two-app model, both apps).

1. **Save.** Provisioning will now **wait** for Connect Before Logon to install and register — and honor

the `3010` reboot — before the user reaches the desktop.

2.5 Apply the AV / IMECache exclusion

Ensure the provisioning-time antivirus policy **excludes** `C:\Program Files (x86)\Microsoft Intune Management Extension\Content` (and the IMECache) from real-time scanning, and that target devices have adequate free disk (R11). This is the single most effective step against the intermittent **"unzipping"** `0x87D30067` failure.

Gate B (Configure): The device group(s) are **Assigned** and **owned by the Provisioning Client SP**; the single Win32 app exists (install + `PanSetup` registry + service-readiness wait + `-registerplap` in SYSTEM + `3010`), is **System** context with the **return codes mapped** and **detection** checking GP≥5.2 + `Prelogon=1` + PLAP registered; the app is an **ESP blocking app**; the AV/IMECache exclusion is applied; and (if cert auth) the certificate profile is assigned. Re-run `GPPLAP-Discovery.ps1` — every expected object should now be found, and the group's SP owner should be confirmed.

Phase 3 — Pilot

1. Take a **Windows 11** test device (physical or VM) registered for Autopilot and in

`PILOT – GlobalProtect PLAP – Autopilot Devices` (confirm the group actually populated — with the SP owner it fills at enrollment).

1. Run a **Hybrid Azure AD Join Autopilot (user-driven)** provisioning from a network position that

mimics production (off the corporate LAN, so Connect Before Logon is genuinely exercised).

1. Let the **Enrollment Status Page** run. It should **block** on

`GlobalProtect – Connect Before Logon (PLAP) – Win32`, install GlobalProtect, apply the registry, register PLAP, take the **ESP-managed reboot** (from the `3010` return), and continue.

1. If ESP fails on the app, collect logs immediately with

`powershell -ExecutionPolicy Bypass -File .\scripts\GPPLAP-Collect-Logs.ps1 -OutFolder C:\Temp\gpplap` and go to `06-troubleshooting.md` before retrying.

Gate C (Pilot): The Windows 11 device completes Hybrid-Join Autopilot with the app *Installed* (the detection rule for GP≥5.2 + `Prelogon=1` + PLAP registered satisfied) and the ESP no longer blocking. If it fails on the unzip error, the missing tile, or a reboot loop, resolve per Troubleshooting before proceeding.

Phase 4 — Validate

1. On the pilot device, run:

`powershell -ExecutionPolicy Bypass -File .\scripts\GPPLAP-Validation.ps1` It confirms GlobalProtect is installed and ≥ **5.2**, `PanSetup\Prelogon = 1` and the portal is set, the **PLAP credential provider is registered**, and reports the device join type via `dsregcmd` (DomainJoined + AzureAdJoined = hybrid).

1. Perform the **decisive** manual test in `05-testing.md`: **lock or sign out** to the Windows 11

sign-in screen, confirm the **Connect Before Logon / network sign-in** tile is **visible**, use it to bring the GlobalProtect tunnel up **before** logon, then perform a **first interactive hybrid domain logon off-network** (a user with **no cached credentials** on this device) — it should succeed over the pre-logon tunnel.

1. Collect a full evidence bundle for the record:

`powershell -ExecutionPolicy Bypass -File .\scripts\GPPLAP-Collect-Logs.ps1 -OutFolder C:\Temp\gpplap`

1. Run the fleet report to see pilot coverage:

`powershell -ExecutionPolicy Bypass -File .\scripts\GPPLAP-Report.ps1`

Gate D (Validate): On the pilot Windows 11 device: the app installs, GP is ≥ **5.2**, `Prelogon=1` with the portal set, **PLAP is registered**, the **Connect Before Logon tile is visible** at the sign-in screen, the **pre-logon tunnel establishes**, the **DC is reachable**, and a **first-time hybrid domain logon off-network succeeds** over the tunnel. All Testing gates are green. Only then proceed.

Phase 5 — Expand

1. **Add** `PROD – GlobalProtect PLAP – Autopilot Devices` (Assigned, SP-owned) to the **Required**

assignment of `GlobalProtect – Connect Before Logon (PLAP) – Win32` (and the certificate profile if used). Keep the pilot group assigned. Ensure the production group is also added to the **ESP blocking app** scope as needed.

1. Expand in waves if the fleet is large (add production sub-groups over several days) and watch the report and Autopilot deployment success between waves.

1. Monitor with `GPPLAP-Report.ps1` for coverage and with Intune **Apps > Monitor** / **Devices > Enrollment > Autopilot deployments** for install and provisioning success.

1. Keep `GPPLAP-Collect-Logs.ps1` on hand for any straggler — it is the fastest path from a failed provisioning to a diagnosed cause.

Rollback (any phase): Remove/exclude the target group from the app assignment and from the ESP blocking-app list; to remove Connect Before Logon, flip the Win32 app assignment to **Uninstall** (the uninstall command can run `PanGPS.exe -unregisterplap` first) and remove any certificate profile.

Gate E (Expand): Fleet coverage meets your target (e.g. ≥ 95% of assigned devices with the app *Installed* and Connect Before Logon registered, provisioning succeeding); remaining failures are triaged. Deployment is complete.

Testing & Validation — GlobalProtect Connect Before Logon (PLAP) for Hybrid Autopilot via Intune

Run these tests on the **pilot Windows 11 device** before expanding, and spot-check during expansion. Each test has an explicit **pass/fail gate**. A phase is only "done" when its gate is **PASS**. Object names match the Implementation guide verbatim.

> ***Scope note:*** *these tests validate that Intune* ***delivered and registered Connect Before Logon during Autopilot*** *and that a device* ***can bring the tunnel up at the sign-in screen and complete a hybrid domain logon over it****. Deep GlobalProtect* ***portal / gateway*** *diagnostics (why a specific portal rejects an auth method, gateway routing) are Palo Alto-side and deferred to Palo Alto docs.*

Test matrix overview

#	Test	Where	Pass gate
T1	Tenant/object readiness + group SP owner	Discovery script	Objects found & assigned; group Assigned + SP owner
T2	Autopilot Hybrid-Join provisioning (Win11)	Autopilot + ESP	Provisioning completes; ESP blocks then releases on the app

#	Test	Where	Pass gate
T3	Win32 app installed + CBL enabled + registered	Intune monitor + device	App *Installed*; GP≥5.2, `Prelogon=1`, PLAP registered
T4	(If cert auth) device certificate applied	Intune monitor + device	Cert *Succeeded*, machine cert in `LocalMachine\My`
T5	Connect Before Logon tile visible at sign-in	Pilot device	CBL / network sign-in tile appears at the lock screen
T6	Pre-logon tunnel + first hybrid logon off-network	Pilot device	Tunnel up before logon; first-time domain logon succeeds
T7	On-device validation script	Pilot device	Script reports PASS
T8	Log-capture runbook	Pilot device	Evidence ZIP produced (IME + Autopilot + GP logs)
T9	Fleet coverage report	Report script	Coverage ≥ target
T10	Rollback / uninstall	Test device	App/PLAP/cert removed; CBL tile gone

T1 — Tenant and object readiness (including the group's SP owner)

Steps: Run `GPPLAP-Discovery.ps1`. Review the readiness table: the app `GlobalProtect – Connect Before Logon (PLAP) – Win32` (or the two-app alternative), the optional cert profile, the device groups `PILOT – / PROD – GlobalProtect PLAP – Autopilot Devices`, whether the app is referenced as an **ESP blocking app**, and — critically — whether the **Assigned** group has the Intune Provisioning Client SP (`f1346770-5b25-470b-88bd-d5744ab7952c`) as an **owner**. **PASS:** Every expected object is present and assigned; the pilot group is **Assigned** and the SP owner is present. **FAIL:** Any object missing/unassigned, or the group missing its SP owner → fix in Requirements/ Implementation (R9 / §2.0) before piloting.

T2 — Autopilot Hybrid-Join provisioning on Windows 11

Steps: Provision a **Windows 11** Autopilot device (user-driven, Hybrid Azure AD Join) from an off-LAN network position. Watch the **Enrollment Status Page**. **PASS:** The ESP **blocks** on `GlobalProtect – Connect Before Logon (PLAP) – Win32`, the app installs, the ESP-managed reboot (from the `3010` return) occurs, and provisioning **completes** to the desktop. **FAIL:** ESP hangs on the app, an "unzipping" error, or a reboot loop → run `GPPLAP-Collect-Logs.ps1` and see Troubleshooting §"Unzipping" / 0x87D30067, §Autopilot ESP hangs on the app, and §Install fails after reboot.

T3 — Win32 app installed, Connect Before Logon enabled and registered

Steps: Intune → **Apps > Monitor > App install status** for `GlobalProtect – Connect Before Logon (PLAP) – Win32`. On the device, confirm GlobalProtect is present and ≥ **5.2**, `HKLM:\SOFTWARE\Palo Alto Networks\GlobalProtect\PanSetup\Prelogon` = `1` with the portal set, and a PLAP credential provider is registered under `HKLM:\SOFTWARE\Microsoft\Windows\CurrentVersion\Authentication\Credential Providers`. **PASS:** Status *Installed* and all three conditions (GP≥5.2, `Prelogon=1`, PLAP registered) are true. **FAIL:** *Failed*/*Pending*, or installed but CBL not registered → see Troubleshooting §PLAP tile not visible and §Install failures.

T4 — Device certificate applied (cert-auth deployments only)

Steps: Confirm `GlobalProtect – PLAP – Device Cert (SCEP) – Windows` shows *Succeeded* in Intune and a **machine** certificate is present in `LocalMachine\My` (`certlm.msc`), chaining to the trusted root. **PASS:** A valid, unexpired **device** certificate is present and chains to a CA the portal trusts. **FAIL:** *Error*/no cert / wrong store (user not machine) → see Troubleshooting §Tunnel up but hybrid first-logon fails (cert/trust) and confirm portal trust with your Palo Alto admin.

T5 — Connect Before Logon tile shows at the sign-in screen

Steps: Lock the Windows 11 device (or sign out) to reach the sign-in screen. Look for the **Connect Before Logon / network sign-in** tile (a credential-provider option that launches the GlobalProtect pre-logon window). A reboot after install may be required for it to appear. **PASS:** The Connect Before Logon / network sign-in tile is visible at the sign-in screen. **FAIL:** No tile → see Troubleshooting §PLAP tile not visible (GP<5.2, `Prelogon` not set, `-registerplap` not run or not in SYSTEM, or registration ran before the GP service was ready).

T6 — Pre-logon tunnel + first hybrid logon off-network (the decisive test)

Steps: From the sign-in screen, use the **Connect Before Logon** tile to establish the GlobalProtect tunnel (portal FQDN from R7) **before** any user signs in. With the tunnel up, perform a **first interactive logon of a hybrid domain user who has never signed in to this device** (no cached credentials), from **off the corporate network**. **PASS:** The tunnel establishes **before** logon; the domain controller is reachable; the first-time hybrid domain logon **completes over the tunnel**. **FAIL:** Tunnel won't establish, or logon fails with the tunnel up → see Troubleshooting §Tunnel up but hybrid first-logon fails (DC/ODJ/time skew) and, for a cached-credential false pass, re-test with a truly new user.

T7 — On-device validation script

Steps: On the pilot device run `GPPLAP-Validation.ps1`. **PASS:** Script reports GlobalProtect installed **and** ≥ **5.2**, `Prelogon=1` with portal set, **and PLAP registered** → overall **PASS** (and the join type shows DomainJoined + AzureAdJoined = hybrid). **FAIL:** Script reports a missing component → address per its message before counting the device.

T8 — Log-capture runbook (the Friday test-window tool)

Steps: On the pilot device run `GPPLAP-Collect-Logs.ps1 -OutFolder C:\Temp\gpplap`. Confirm it produces a timestamped ZIP containing the IME logs (`IntuneManagementExtension.log`, `AgentExecutor.log`, `AppWorkload.log`), the `MdmDiagnosticsTool.exe -area`

`Autopilot;DeviceEnrollment;DeviceProvisioning -zip` output, the GlobalProtect PanGPS/PanGPA logs, and the relevant Event Viewer channels. **PASS:** A single evidence ZIP is produced containing the IME, Autopilot/MDM, and GlobalProtect logs. **FAIL:** Missing sections → re-run elevated; confirm the source paths exist on this device.

T9 — Fleet coverage report

Steps: Run `GPPLAP-Report.ps1`; review app install coverage (and cert coverage if used) across the Autopilot device groups. **PASS (pilot):** 100% of pilot devices *Installed* with Connect Before Logon registered. **PASS (expand):** Coverage ≥ your target (e.g. ≥ 95%) with remaining failures triaged. **FAIL:** Coverage below target → remediate (T2/T3 failures) before widening further.

T10 — Rollback / uninstall verification

Steps: On a disposable test device, exclude it from the target group (or flip the Win32 app to **Uninstall**). Sync and reboot. **PASS:** GlobalProtect and the PLAP registration are removed (the Connect Before Logon tile disappears), the certificate (if used) is removed, and the device returns to its pre-deployment state. **FAIL:** Residual app / CBL tile / cert → confirm the Uninstall command (incl. `-unregisterplap`) and the unassignment took effect.

Overall exit criteria (before Expand)

All of T1–T8 **PASS** on the pilot Windows 11 device (including the decisive pre-logon tunnel + first hybrid logon test T6), T9 shows 100% pilot coverage, and T10 confirms a clean rollback. Record results against each gate; only then proceed to Phase 5 (Expand).

Troubleshooting — GlobalProtect Connect Before Logon (PLAP) for Hybrid Autopilot via Intune

Common failures and fixes, grouped by symptom. Object names match the Implementation guide. Work top-down within a section — the first cause listed is the most common. On **any** failed provisioning, run `GPPLAP-Collect-Logs.ps1 -OutFolder C:\Temp\gpplap` **before** wiping the device — a wipe destroys the evidence you need.

> ***Vendor boundary:*** *problems on the* ***GlobalProtect portal / gateway*** *(portal rejects the auth method, gateway routing, head-end CA trust) are* ***deferred to Palo Alto Networks' official documentation and your Palo Alto admin****. This guide resolves the* ***Intune / Autopilot / Windows delivery*** *side and tells you when a failure points at the Palo Alto side.*

§"Unzipping" / 0x87D30067 — the Win32 app fails to unzip during Autopilot

Symptom: `GlobalProtect – Connect Before Logon (PLAP) – Win32` fails during the Enrollment Status Page with an **"unzipping"** error; the IME log shows `0x87D30067`. The IME downloaded the `.intunewin` but could not extract it into its content/cache path.

1. **Antivirus is locking or scanning the IMECache during extraction.** This is the most common cause.

Microsoft Defender or a third-party AV holds a lock on `C:\Program Files (x86)\Microsoft Intune Management Extension\Content` (the IMECache) while the IME tries to unzip. **Add that path (and the IMECache) to the AV exclusion list applied during provisioning** (R11) and re-provision.

1. **Low disk space.** Extraction needs room for the unzipped payload. Confirm the device has adequate

free disk; clear space and retry.

1. **Corrupt or improperly re-wrapped `.intunewin`.** A package wrapped from an **already-wrapped**

payload (nesting), or wrapped with a stale tool, extracts inconsistently. **Re-wrap cleanly** with the **current Microsoft Win32 Content Prep Tool** (R12) from a minimal source folder (the MSI + the script) — never nest an already-wrapped file.

1. **Path / permission tampering.** A hardened image or a GPO that alters permissions on the IME content

path can block extraction. Confirm the IME content path exists with default permissions.

1. **Verify the source folder.** Confirm the MSI and the install script are actually present in the

source folder you wrapped, and the install command matches the script filename.

Confirm the fix: `GPPLAP-Collect-Logs.ps1` flags `0x87D30067` in the IME log; after the AV exclusion and a clean re-wrap, the app installs and the flag is gone.

§PLAP tile not visible — no Connect Before Logon / network sign-in at the sign-in screen

Symptom: Provisioning finishes but the **Connect Before Logon / network sign-in** tile never appears at the Windows 11 sign-in screen, so there is no way to bring the tunnel up before logon.

1. **GlobalProtect app is below 5.2.** Connect Before Logon does not exist before app **5.2**. Confirm the

installed version with `GPPLAP-Validation.ps1`; if it is below 5.2, package the correct MSI (R6) and redeploy.

1. `Prelogon` **registry not set.** The PLAP provider stays dormant unless

`HKLM:\SOFTWARE\Palo Alto Networks\GlobalProtect\PanSetup\Prelogon` = `1` (with a `Portal` value under the same key). Confirm with `GPPLAP-Validation.ps1`; if missing, fix the install script to write the `PanSetup` registry **before** registration and redeploy.

1. `PanGPS.exe -registerplap` **was never run, or not in the SYSTEM context.** Registration is what adds

the tile. It must run as **SYSTEM** (the Win32 app runs as System). If the app ran as user, or the registration step was skipped, the provider is not registered — confirm the credential-provider registration under `HKLM:\SOFTWARE\Microsoft\Windows\CurrentVersion\Authentication\Credential Providers` and set the app's **Install behavior = System**, then redeploy.

1. **Registration ran before the GlobalProtect service (PanGPS) was ready.** If `-registerplap` executed

before `PanGPS` was *Running*, registration silently no-ops. Add the **service-readiness wait** to the install script (poll `Get-Service PanGPS` until *Running*, bounded) so registration always runs after the service exists; redeploy.

1. **No reboot after install.** The tile may only surface after a reboot. Ensure the app **returns 3010**

so the ESP takes the reboot (see the next section) and re-check at the sign-in screen.

§Install fails after reboot / IME state lost — a reboot loop or lost app state

Symptom: The app "installs" then the device reboots and the install is lost, retried, or the ESP loops; the IME log shows the agent restarting mid-install.

1. **The install script called a forced** `Restart-Computer`**.** This is the root cause to eliminate. A

scripted reboot during the ESP interrupts the Intune Management Extension and loses app state. **Remove any** `Restart-Computer` **from the script** and instead `exit 3010` so **Intune/ESP owns the reboot**.

1. **Return codes not mapped on the app.** If the app does not know `3010` means "soft reboot", it treats

the reboot request as a failure. On the Win32 app, map return codes: `0`/`1707` success, `3010` **soft reboot**, `1618` retry.

1. **Missing** `Start-Process … -Wait`**.** Without `-Wait`, the MSI (or `-registerplap`) returns immediately

and the app reports complete before the work finishes — the reboot then lands mid-install. Ensure the MSI and the registration are launched with `-Wait`.

1. **Device restart behavior override.** Set the app's **Device restart behavior** to *Determine behavior

based on return codes* so the `3010` mapping actually drives the reboot.

§Autopilot ESP hangs on the app — provisioning blocks and never releases

Symptom: The Enrollment Status Page sits on `GlobalProtect – Connect Before Logon (PLAP) – Win32` and never completes.

1. **The app genuinely failed (unzip / install error).** Check the IME log via `GPPLAP-Collect-Logs.ps1`;

resolve per §"Unzipping" / 0x87D30067 or §Install fails after reboot.

1. **Detection rule never becomes true.** If detection checks something the install never writes, the app

never reports *Installed* and the ESP blocks forever. Detection must check **GP≥5.2 AND** `Prelogon=1` **AND the PLAP credential-provider registration** — and each of those must actually be produced by the install. Fix the detection rule to match what the script writes.

1. **The app returns an unmapped non-zero code.** If the script exits with a code the app treats as

failure (and it is not `3010`/`1618`), the ESP fails. Confirm the script's exit codes and the app's return-code mapping.

1. **App not correctly on the ESP blocking list.** Confirm the app is listed under the ESP's blocking apps

(`ESP – Autopilot – GlobalProtect PLAP Blocking`) and that the device's group is in the ESP's scope.

§Enrollment-time device group empty — nothing deploys during Autopilot

Symptom: The device provisions but the GlobalProtect PLAP app (and any profile) never targets it; the Autopilot **device group is empty at enrollment** even though you added devices.

1. **The group is missing the Provisioning Client SP owner.** Enrollment-time device grouping only works

when the **Assigned** group is **owned** by the Intune Provisioning Client service principal (**AppId** `f1346770-5b25-470b-88bd-d5744ab7952c`). Add the SP as an **owner**: Entra → **Groups > <group> > Owners > Add owners** → search by **AppId** (the display name, often *Intune Autopilot ConfidentialClient*, varies). `GPPLAP-Discovery.ps1` flags a missing SP owner; `GPPLAP-Set-ScopeTargeting.ps1` sets it.

1. **The group is Dynamic, not Assigned.** Enrollment-time grouping needs an **Assigned** group. If the group was created as Dynamic, recreate it as **Assigned** and re-add devices + the SP owner.

1. **The SP is absent from the tenant.** If the Provisioning Client SP cannot be found, provision it by opening the Intune portal **once as a Global Administrator**, then add it as owner.

§Works on Windows 10 but not Windows 11 (or vice-versa)

Symptom: The exact same package that behaved on the Windows 10 test machine behaves differently on the Windows 11 production target (tile missing, registration timing, or install result differs).

1. **You validated on the wrong OS.** Early testing here was on Windows 10; **production is Windows 11**.

Do not assume parity — **re-test the full flow on Windows 11** (pilot device, `GPPLAP-Validation.ps1`, and the sign-in-tile check). Make Windows 11 the pilot OS.

1. **OS-version requirement mismatch on the app.** Confirm the app's **Requirements** minimum OS matches

the Windows 11 target and Palo Alto's support matrix, so the app is *Applicable* on Windows 11.

1. **Service-readiness timing differs across OS builds.** If registration timing is marginal, the difference surfaces as "works on one OS, not the other." Ensure the **service-readiness wait** is present (not a fixed sleep) so registration reliably follows `PanGPS` becoming *Running* on either OS.

1. **If a Windows 10 estate remains,** treat it as a **separate target** and re-validate there rather than shipping the Windows-11-validated package blindly.

§Tunnel up but hybrid first-logon fails — Connect Before Logon connects yet Windows sign-in doesn't work

Symptom: The Connect Before Logon tunnel establishes at the sign-in screen, but the subsequent first-time **hybrid domain** logon fails ("no logon servers available", first-time logon rejected, or password

change won't complete).

1. **Domain controller not reachable over the tunnel.** The tunnel is up but the DC isn't routed. Confirm

with your Palo Alto/network team that the **DC subnets are inside the tunnel** and reachable from the VPN pool. Split-tunnel routing that excludes DC subnets is a common cause (Palo Alto-side).

1. **On-prem Domain Join (ODJ) not completing.** Hybrid Azure AD Join Autopilot completes domain join via

the **Intune Connector for Active Directory**. Confirm the connector is **Active** (R4) and the **Domain Join** profile targets the right domain/OU; a device that can reach the DC but has no ODJ blob won't finish join.

1. **DNS not resolving internal names pre-logon.** The pre-logon tunnel must hand the client internal DNS

so the DC name resolves. Confirm DNS pushed by the portal/gateway (Palo Alto-side) is applied before logon.

1. **Time skew.** A large clock difference between the device and the DC breaks Kerberos/domain logon.

Confirm the device time is correct (Autopilot devices off-network can drift before first sync).

1. **Certificate rejected by the portal (cert auth).** If Connect Before Logon uses a certificate and the

tunnel authenticates but access is denied, the **portal must trust the issuing CA** — a **Palo Alto-side** trust decision, deferred to Palo Alto docs. Provide your Palo Alto admin the issuing CA chain; confirm the device cert is a **machine** cert in `LocalMachine\My` (T4).

1. **Cached-credential test masks the issue.** If you tested with a user who has signed in before, Windows

may use cached credentials rather than the tunnel. **Re-test with a first-time / no-cached-credential logon** to truly exercise pre-logon reachability.

§Certificate issues — device certificate missing, wrong store, or rejected (cert-auth deployments)

Symptom: `GlobalProtect – PLAP – Device Cert (SCEP) – Windows` fails, or the pre-logon tunnel can't authenticate.

1. **SCEP profile not applied yet.** Confirm the profile shows *Succeeded* in Intune and the device has

synced. `GPPLAP-Discovery.ps1` shows whether it is assigned to the pilot device group.

1. **Certificate issued to the user store, not the machine store.** Connect Before Logon needs a **Device

(machine) **certificate in** `LocalMachine\My` **(there is no user context at the sign-in screen). If** Certificate type **was *User*, set it to** Device** and redeploy.

1. **Trusted root not deployed first.** The device must trust the issuing CA before the SCEP profile

succeeds. Confirm the trusted-root profile is *Succeeded*, then the SCEP profile.

1. **Head-end doesn't trust the issuing CA (Palo Alto side).** If the cert is issued correctly but the

portal still rejects it, the **portal/gateway must trust the issuing CA** — Palo Alto-side, deferred to Palo Alto docs.

§Reporting / script issues

1. `Connect-MgGraph` **fails or lacks scopes.** Install the SDK (`Install-Module Microsoft.Graph -Scope CurrentUser) and consent to the scopes each script lists in its header. Read-only scripts need only .Read.All` scopes.

1. **Objects show MISSING but exist in the portal.** The scripts match **by exact display name** — confirm

the object is named to convention verbatim (including the en-dash – and spacing). Rename any drifted object to the convention.

1. **Discovery flags the group's SP owner as missing but you added it.** Confirm you added the SP by

AppId `f1346770-5b25-470b-88bd-d5744ab7952c` (the display name varies) and that the group is **Assigned**, not Dynamic.

1. **Validation says PLAP not registered but the tile shows.** The script checks the credential-provider

registration and the GP service/DLL; if Palo Alto changed the program path/DLL name for your version, note the actual path (per Palo Alto docs) — the **logon-screen tile is the ground truth**.

1. **Log collector missing sections.** Run `GPPLAP-Collect-Logs.ps1` **elevated** so the IME logs, MDM

diagnostics, and Event channels can be read; confirm the source paths exist on this device.

If a failure persists after the Intune / Autopilot / Windows side is confirmed healthy (validation PASS, app *Installed*, `Prelogon=1`, PLAP registered, the Connect Before Logon tile showing, and the tunnel establishing) and the **hybrid logon still fails or the portal still rejects the connection**, treat it as a **GlobalProtect portal / gateway issue** and continue with **Palo Alto Networks' official documentation** and your Palo Alto admin.

Best Practices — GlobalProtect Connect Before Logon (PLAP) for Hybrid Autopilot via Intune

Field-proven practices for the deploy-and-diagnose model. These make the Autopilot build predictable, reversible, and easy to troubleshoot when the enrollment sequence misbehaves.

Naming and structure

• **Name every object to the convention and reuse it verbatim.** The app, the cert profile, the device

groups, the ESP reference, and the filter all follow the Implementation table (`GlobalProtect – Connect Before Logon (PLAP) – Win32`, `PILOT – GlobalProtect PLAP – Autopilot Devices`, …). The discovery, validation, log-capture, and report scripts find objects **by these names** — drift breaks automation and reporting (including the en-dash – and spacing).

- **Prefer the single Win32 app.** Keep install, the `PanSetup` registry, the service-readiness wait, and

`PanGPS.exe -registerplap` in **one** app. Splitting them into a two-app model reintroduces the ordering/reboot seam where the unzip and sequencing failures cluster — only split when your environment already does.

- **Keep pilot and production device groups separate and both assigned.**

`PILOT – GlobalProtect PLAP – Autopilot Devices` stays assigned after expansion so you retain a fast Windows 11 test cohort.

Autopilot enrollment-time targeting

- **Use an Assigned group owned by the Provisioning Client SP.** Enrollment-time device grouping only

works when the group is **Assigned** (not Dynamic) and the Intune Provisioning Client service principal (**AppId** `f1346770-5b25-470b-88bd-d5744ab7952c`) is an **owner**. Without the owner the group is empty at enrollment and nothing deploys — verify the owner in discovery, not just the group.

- **Make the app an ESP blocking app.** So provisioning **waits** for Connect Before Logon to install and

register before the user reaches the desktop, and honors the app's `3010` reboot at the right moment.

Reboot sequencing (the rule that removes the biggest failure)

- **Never call** `Restart-Computer` **from the install script during the ESP.** It interrupts the Intune

Management Extension and causes install failures / lost state. **Return exit code** `3010` and let **Intune/ESP** own the reboot.

- **Map the return codes on the app:** `0`/`1707` = success, `3010` = soft reboot, `1618` = retry.

- **Always** `Start-Process … -Wait` for the MSI (and for `-registerplap`) so the app never reports

complete before the installer/registration actually finishes.

Avoiding the "unzipping" 0x87D30067 failure

- **Exclude the IME content/IMECache path from AV during provisioning**

(`C:\Program Files (x86)\Microsoft Intune Management Extension\Content` and the IMECache). AV locking or scanning the cache during extraction is the most common cause.

- **Re-wrap cleanly with the current Content Prep Tool.** Never nest an already-wrapped payload; keep the

source folder minimal (MSI + script). Keep enough free disk on target devices.

Connect Before Logon correctness

- **Require GlobalProtect ≥ 5.2.** Connect Before Logon does not exist below 5.2 — enforce it in the

app's detection rule, not just the requirements.

- **Set the** `PanSetup` **registry before registration and before the sign-in screen.** `Prelogon = 1` and

the default `Portal` must exist before `-registerplap` and before the lock screen is reached.

- **Register PLAP as SYSTEM, after the service is ready.** Run `PanGPS.exe -registerplap` in the System

context (the Win32 app runs as System), and only after `PanGPS` is *Running* — add a bounded readiness wait rather than a fixed sleep.

- **Detect all three conditions.** Detection should check GP≥5.2 **and** `Prelogon=1` **and** the PLAP

credential-provider registration — detecting only the MSI hides a device where the tile will never appear.

Windows 11 vs Windows 10

- **Pilot and validate on the production OS (Windows 11).** "Works on Win10, not Win11" (or the reverse)

is a real early-signal in this scenario — never assume parity. Re-test the full flow on Windows 11 and, if a Windows 10 estate remains, test it as a separate target.

Diagnose fast

- **Keep** `GPPLAP-Collect-Logs.ps1` **in the test-window kit.** On any failed provisioning, capture the IME

logs (`IntuneManagementExtension.log`, `AgentExecutor.log`, `AppWorkload.log`), the `MdmDiagnosticsTool.exe -area Autopilot;DeviceEnrollment;DeviceProvisioning -zip` bundle, and the GlobalProtect `PanGPS`/`PanGPA` logs **before** wiping and retrying — a wipe destroys the evidence.

- **Read the IME log for the app exit code.** The `3010`/`1618`/error the app returned, and the unzip

`0x87D30067` if present, are in `IntuneManagementExtension.log` / `AppWorkload.log`.

Certificates (if cert auth)

- **Use a Device (machine) certificate.** There is no user context at the sign-in screen; issue the cert

to `LocalMachine\My` and deploy the trusted root **first**, then the SCEP profile.

- **Ensure the portal trusts the issuing CA** (Palo Alto-side trust config per Palo Alto docs) and set a

renewal threshold comfortably before expiry so Connect Before Logon never silently loses auth.

Change control and rollback

- **Keep a one-click rollback ready:** exclude the group, remove the app from the ESP blocking list, or

flip the Win32 app to **Uninstall** (running `PanGPS.exe -unregisterplap` first). Test the rollback (Testing T10) before you ever need it.

- **Deploy inside a change window** and announce it — this changes the provisioning and sign-in

experience for real users.

• **Version your inputs.** Track which GlobalProtect MSI version, portal FQDN, and auth method a given rollout used, so you can reproduce or revert precisely.

Security hygiene

• **Never embed VPN credentials.** Use the portal's certificate or SAML/credential prompt at the Connect Before Logon window; do not store secrets in the registry or the package.

• **Least-privilege admin.** Use Intune Administrator + Groups Administrator for the operator; a Global Administrator may be needed once to provision/own the first-party SP on the group.

• **Keep the client patched.** Update the app package when Palo Alto releases fixes; pilot the new version on Windows 11 before fleet-wide expansion, same as any change.

Deliverables Summary — GlobalProtect Connect Before Logon (PLAP) for Hybrid Autopilot via Intune

This solution is a complete, self-guided package to deploy **GlobalProtect Connect Before Logon (PLAP)** during a **Hybrid Azure AD Join Windows Autopilot** provisioning using Microsoft Intune — pilot-first, rollback-ready, and built to be **diagnosed** as well as deployed. Below is everything included.

Documentation set (source/)

Document	What it gives you
`01-solution-overview.md`	The scenario (Always On → user-initiated VPN after a security incident; Hybrid Autopilot needs pre-logon DC reachability), what Connect Before Logon (CBL/PLAP) is, the single-Win32-app delivery model (install + `PanSetup` registry + service-readiness wait + `PanGPS.exe -registerplap` + return `3010`), the Autopilot/ESP fit, and the Assess→Configure→Pilot→Validate→Expand model.
`02-requirements.md`	Every prerequisite with **Description / Why / Verify / Implement-if-missing** — Intune licensing + Autopilot, the Hybrid-Join profile, Domain Join + ODJ connector, ESP, the GlobalProtect 5.2+ MSI + portal FQDN + auth method, optional PKI, the **Assigned** device group **owned by the Provisioning Client SP**, reboot-sequencing design, and the AV/IMECache exclusion. Includes the Government note.

Document	What it gives you
`03-implementation.md`	Exact ordered build with **Gates A–E**: create the Assigned + SP-owned device group, build the single Win32 app (install + registry + readiness wait + `-registerplap` in SYSTEM + `3010`), set System context, detection (GP≥5.2 + `Prelogon=1` + PLAP registered) and return-code mapping, add the app as an **ESP blocking app**, apply the AV/IMECache exclusion, then pilot on Windows 11, validate, and expand. Documents the two-app alternative.
`05-testing.md`	The validation test matrix (T1–T10) with explicit **PASS/FAIL** gates, including the decisive pre-logon tunnel + first hybrid domain logon test and the log-capture runbook.
`06-troubleshooting.md`	The failure-mode matrix: the **"unzipping" / 0x87D30067** error, the **PLAP tile not visible**, **install fails after reboot / IME state lost**, **ESP hangs on the app**, **Win10-vs-Win11** mismatch, **tunnel up but hybrid first-logon fails**, and **enrollment-time group empty**.
`07-best-practices.md`	Field-proven practices for naming, enrollment-time targeting, reboot sequencing, avoiding the unzip failure, Connect Before Logon correctness, Windows 11 testing, fast diagnosis, certificates, and safe operation.
`executive-summary.md`	One-page business-outcome summary for stakeholders.
`global-requirements-network.md`	Category-level Network prerequisites (shared across Network solutions).

PowerShell scripts (scripts/)

Discovery/report/scope-targeting use the **Microsoft Graph PowerShell SDK**; validation and log-capture run **locally** on the device and need no Graph. Discovery/validation/report/log-capture are **read-only / collection-only**; scope-targeting **defaults to** `-WhatIf` **(preview)** and changes nothing until you re-run with `-WhatIf:$false`. Each script's header documents its exact purpose, run command, and required Graph scopes.

Script	Purpose	Mode
`GPPLAP-Discovery.ps1`	Readiness sweep across Entra + Intune — finds the Win32 app (and the two-app alternative names), the optional cert profile, the pilot/prod Autopilot **device** groups, flags whether the **Assigned** group has the Provisioning Client SP (`f1346770-5b25-470b-88bd-d5744ab7952c`) as **owner**, and whether the app is referenced as an **ESP blocking app**.	Read-only
`GPPLAP-Validation.ps1`	Confirms **on this Windows 11 device** that GlobalProtect is installed and ≥ **5.2**, `PanSetup\Prelogon = 1` with the portal set, the **PLAP credential provider is registered**, and reports join type via `dsregcmd`. Prints an overall PASS/FAIL.	Read-only (local)
`GPPLAP-Collect-Logs.ps1`	The test-window diagnostic collector — gathers the IME logs, runs `MdmDiagnosticsTool.exe -area Autopilot;DeviceEnrollment;DeviceProvisioning -zip`, collects GlobalProtect `PanGPS`/`PanGPA` logs and Event Viewer channels, and ZIPs everything to a timestamped folder.	Collection-only (local)
`GPPLAP-Report.ps1`	Fleet coverage report — per-app install counts (installed/failed/pending) and the optional cert profile across the Autopilot device groups, with coverage percentages.	Read-only
`GPPLAP-Set-ScopeTargeting.ps1`	Creates/populates the pilot (and optional production) Autopilot **device** groups as **Assigned** security groups, adds devices by name, **and sets the Provisioning Client SP as an owner** (required for enrollment-time targeting).	Write, **default** `-WhatIf`

The consistent object naming convention

Every artifact is named to a convention and reused verbatim across docs, tests, troubleshooting, and scripts (so the scripts find objects deterministically):

- Device groups: `PILOT – GlobalProtect PLAP – Autopilot Devices`, `PROD – GlobalProtect PLAP – Autopilot Devices`
- Win32 app (recommended): `GlobalProtect – Connect Before Logon (PLAP) – Win32`
- Two-app alternative: `GlobalProtect – VPN Client – Win32` + `GlobalProtect – PLAP Registration – Win32`
- Optional certificate profile: `GlobalProtect – PLAP – Device Cert (SCEP) – Windows`
- ESP blocking-app reference: `ESP – Autopilot – GlobalProtect PLAP Blocking`

Vendor boundary (what is deferred)

The **GlobalProtect portal / gateway configuration, the authentication method the portal enforces, and the GlobalProtect app internals are deferred to Palo Alto Networks' official documentation.** This solution authors the **Intune, Autopilot, Entra, and Windows** side at full click-by-click detail and consumes the Palo Alto-supplied **MSI (5.2+)**, **portal FQDN**, and **auth method** as inputs.

Appendix A — PowerShell Scripts

Included in the accompanying ZIP. Each script's header documents full usage.

GPPLAP-Collect-Logs.ps1

Purpose: Local, collection-only diagnostic runbook for a Windows 11 Autopilot test device. Gathers, into a single timestamped ZIP, everything needed to diagnose a Connect Before Logon / Autopilot failure — the "unzipping" (0x87D30067) error, a missing sign-in tile, a reboot loop, or a hybrid first-logon failure — BEFORE the device is wiped and the evidence is lost: - Intune Management Extension logs (C:\ProgramData\Microsoft\IntuneManagementExtension\Logs* — IntuneManagementExtension.log, AgentExecutor.log, AppWorkload.log, and siblings) - Autopilot / MDM diagnostics via MdmDiagnosticsTool.exe -area Autopilot;DeviceEnrollment;DeviceProvisioning -zip - GlobalProtect PanGPS / PanGPA logs from the GlobalProtect install / ProgramData directories - Relevant Event Viewer channels (application/system + MDM/DeviceManagement providers) - A snapshot of the PanSetup registry and the app install exit codes seen in the IME log Makes NO configuration changes — it only reads and copies. HOW TO RUN powershell -ExecutionPolicy Bypass -File .\GPPLAP-Collect-Logs.ps1 -OutFolder C:\Temp\gpplap Run ELEVATED (SYSTEM/admin) so the IME logs, MDM diagnostics, and Event logs can be read. No Microsoft Graph scopes needed — this inspects the local device only. EXPECTED RESULTS A ZIP at <OutFolder>\GPPLAP-Logs-<COMPUTERNAME>-<timestamp>.zip containing the IME logs, the MDM diagnostics ZIP, the GlobalProtect logs, the exported Event Viewer channels, and a summary.txt with the detected app exit codes and PanSetup state. Hand this ZIP to whoever is diagnosing the failure. .NOTES White-label — the repeatable way. Read-only/collection only. GlobalProtect portal/gateway and app internals deferred to vendor docs. >

How to run: powershell -ExecutionPolicy Bypass -File .\GPPLAP-Collect-Logs.ps1 -OutFolder C:\Temp\gpplap Run ELEVATED (SYSTEM/admin) so the IME logs, MDM diagnostics, and Event logs can be read. No Microsoft Graph scopes needed — this inspects the local device only. EXPECTED RESULTS A ZIP at <OutFolder>\GPPLAP-Logs-<COMPUTERNAME>-<timestamp>.zip containing the IME logs, the MDM diagnostics ZIP, the GlobalProtect logs, the exported Event Viewer channels, and a summary.txt with the detected app exit codes and PanSetup state. Hand this ZIP to whoever is diagnosing the failure. .NOTES White-label — the repeatable way. Read-only/collection only. GlobalProtect portal/gateway and app internals deferred to vendor docs. >

GPPLAP-Discovery.ps1

Purpose: Read-only readiness sweep for a GlobalProtect Connect Before Logon (PLAP) deployment during Hybrid Azure AD Join Windows Autopilot via Intune. Finds the single Win32 app 'GlobalProtect – Connect Before Logon (PLAP) – Win32' (and the two-app alternative names), the optional SCEP certificate profile, and the pilot/production Autopilot DEVICE groups. Flags whether each Assigned group has the Intune Provisioning Client service principal (AppId f1346770-5b25-470b-88bd-d5744ab7952c) as an OWNER — required for enrollment-time targeting — and whether the app is referenced as a blocking app on any Enrollment Status Page. Makes NO changes. HOW TO RUN powershell -ExecutionPolicy Bypass -File .\GPPLAP-Discovery.ps1 Required Microsoft Graph scopes (delegated, read-only): DeviceManagementApps.Read.All DeviceManagementConfiguration.Read.All DeviceManagementServiceConfig.Read.All Group.Read.All Directory.Read.All EXPECTED RESULTS A readiness table: the Win32 app Found/Missing with assignment state, the optional cert profile, each Autopilot device group Found/Missing with membership type (Assigned) and SP-owner status, and

whether the app is an ESP blocking app. All green = ready to pilot; any Missing / Dynamic group / missing SP owner / not-blocking = fix before deploying (see 02-requirements.md / 03-implementation.md). .NOTES White-label — the repeatable way. Read-only against Microsoft Graph. GlobalProtect portal/gateway and app internals deferred to vendor docs. >

How to run: powershell -ExecutionPolicy Bypass -File .\GPPLAP-Discovery.ps1 Required Microsoft Graph scopes (delegated, read-only): DeviceManagementApps.Read.All DeviceManagementConfiguration.Read.All DeviceManagementServiceConfig.Read.All Group.Read.All Directory.Read.All EXPECTED RESULTS A readiness table: the Win32 app Found/Missing with assignment state, the optional cert profile, each Autopilot device group Found/Missing with membership type (Assigned) and SP-owner status, and whether the app is an ESP blocking app. All green = ready to pilot; any Missing / Dynamic group / missing SP owner / not-blocking = fix before deploying (see 02-requirements.md / 03-implementation.md). .NOTES White-label — the repeatable way. Read-only against Microsoft Graph. GlobalProtect portal/gateway and app internals deferred to vendor docs. >

GPPLAP-Report.ps1

Purpose: Read-only fleet coverage report for the GlobalProtect Connect Before Logon (PLAP) Autopilot deployment. For the single Win32 app 'GlobalProtect – Connect Before Logon (PLAP) – Win32' (and the two-app alternative, if used) and the optional SCEP certificate profile, pulls Intune install/apply status counts (installed, failed, pending, not applicable) and prints a coverage summary plus an optional CSV. Use it to prove pilot coverage and to watch drift during expansion. Makes NO changes. HOW TO RUN powershell -ExecutionPolicy Bypass -File .\GPPLAP-Report.ps1 Optional: -CsvPath .\gpplap-coverage.csv to export per-object rows. Required Microsoft Graph scopes (delegated, read-only): DeviceManagementApps.Read.All DeviceManagementConfiguration.Read.All DeviceManagementManagedDevices.Read.All EXPECTED RESULTS A table per app/profile with Installed/Succeeded, Failed, Pending, NotApplicable counts and a coverage percentage, plus an overall coverage figure. Target: pilot 100%; expansion >= your threshold (e.g. 95%). Rows with Failed>0 flag where to run GPPLAP-Collect-Logs.ps1 on the device. .NOTES White-label — the repeatable way. Read-only against Microsoft Graph. GlobalProtect portal/gateway and app internals deferred to vendor docs. >

How to run: powershell -ExecutionPolicy Bypass -File .\GPPLAP-Report.ps1 Optional: -CsvPath .\gpplap-coverage.csv to export per-object rows. Required Microsoft Graph scopes (delegated, read-only): DeviceManagementApps.Read.All DeviceManagementConfiguration.Read.All DeviceManagementManagedDevices.Read.All EXPECTED RESULTS A table per app/profile with Installed/Succeeded, Failed, Pending, NotApplicable counts and a coverage percentage, plus an overall coverage figure. Target: pilot 100%; expansion >= your threshold (e.g. 95%). Rows with Failed>0 flag where to run GPPLAP-Collect-Logs.ps1 on the device. .NOTES White-label — the repeatable way. Read-only against Microsoft Graph. GlobalProtect portal/gateway and app internals deferred to vendor docs. >

GPPLAP-Set-ScopeTargeting.ps1

Purpose: Creates the enrollment-time assignment target groups for the GlobalProtect Connect Before Logon (PLAP) Autopilot deployment — 'PILOT – GlobalProtect PLAP – Autopilot Devices' and (optionally) 'PROD – GlobalProtect PLAP – Autopilot Devices' — as ASSIGNED security groups (NOT dynamic), adds the DEVICES you supply (by device display name), AND sets the Intune Provisioning Client first-party service principal (AppId f1346770-5b25-470b-88bd-d5744ab7952c) as an OWNER of each group. The SP

owner is what makes enrollment-time device grouping work: without it the Assigned group stays empty at enrollment and nothing (apps/profiles) deploys during Autopilot. Idempotent: an existing group is reused, members/owner already present are skipped. Defaults to preview only (-WhatIf); re-run with -WhatIf:$false to actually create/populate. HOW TO RUN Preview: powershell -ExecutionPolicy Bypass -File .\GPPLAP-Set-ScopeTargeting.ps1 -PilotDevices DEVICE01,DEVICE02 Apply: powershell -ExecutionPolicy Bypass -File .\GPPLAP-Set-ScopeTargeting.ps1 -PilotDevices DEVICE01 -WhatIf:$false Required Microsoft Graph scopes (delegated): Group.ReadWrite.All GroupMember.ReadWrite.All Device.Read.All Application.Read.All Directory.Read.All EXPECTED RESULTS The pilot group (and prod group if -CreateProd) exist as ASSIGNED security groups, contain the listed devices, and have the Provisioning Client SP as an owner. In preview each create/add is prefixed [WHATIF]; with -WhatIf:$false each prints [DONE]/[FAILED]. These exact group names are the enrollment-time assignment targets in 03-implementation.md. .NOTES White-label — the repeatable way. Write actions gated behind -WhatIf (default ON). GlobalProtect portal/gateway and app internals deferred to vendor docs. >

How to run: Preview: powershell -ExecutionPolicy Bypass -File .\GPPLAP-Set-ScopeTargeting.ps1 -PilotDevices DEVICE01,DEVICE02 Apply: powershell -ExecutionPolicy Bypass -File .\GPPLAP-Set-ScopeTargeting.ps1 -PilotDevices DEVICE01 -WhatIf:$false Required Microsoft Graph scopes (delegated): Group.ReadWrite.All GroupMember.ReadWrite.All Device.Read.All Application.Read.All Directory.Read.All EXPECTED RESULTS The pilot group (and prod group if -CreateProd) exist as ASSIGNED security groups, contain the listed devices, and have the Provisioning Client SP as an owner. In preview each create/add is prefixed [WHATIF]; with -WhatIf:$false each prints [DONE]/[FAILED]. These exact group names are the enrollment-time assignment targets in 03-implementation.md. .NOTES White-label — the repeatable way. Write actions gated behind -WhatIf (default ON). GlobalProtect portal/gateway and app internals deferred to vendor docs. >

GPPLAP-Validation.ps1

Purpose: Local, read-only validation run ON a pilot Windows 11 device. Confirms GlobalProtect is installed and at version 5.2 or later, the PanSetup registry enables Connect Before Logon (Prelogon = 1) with a Portal set, the PLAP credential provider is registered, and reports the device join type via dsregcmd (DomainJoined + AzureAdJoined = Hybrid Azure AD Join). Prints a clear PASS/FAIL so a pilot device can be counted as validated. Makes NO changes. HOW TO RUN powershell -ExecutionPolicy Bypass -File .\GPPLAP-Validation.ps1 No Microsoft Graph scopes needed — this inspects the local device only. Run elevated for the most reliable registry/service reads. EXPECTED RESULTS A checklist: GlobalProtect installed (yes/no + version, >=5.2), PanSetup Prelogon=1 (yes/no), Portal set (value), PLAP credential provider registered (yes/no), and the dsregcmd join type. Overall PASS when GP >= 5.2 AND Prelogon = 1 AND PLAP is registered; otherwise FAIL with the missing item. .NOTES White-label — the repeatable way. Read-only local inspection. GlobalProtect portal/gateway and app internals deferred to vendor docs. >

How to run: powershell -ExecutionPolicy Bypass -File .\GPPLAP-Validation.ps1 No Microsoft Graph scopes needed — this inspects the local device only. Run elevated for the most reliable registry/service reads. EXPECTED RESULTS A checklist: GlobalProtect installed (yes/no + version, >=5.2), PanSetup Prelogon=1 (yes/no), Portal set (value), PLAP credential provider registered (yes/no), and the dsregcmd join type. Overall PASS when GP >= 5.2 AND Prelogon = 1 AND PLAP is registered; otherwise FAIL with the missing item. .NOTES White-label — the repeatable way. Read-only local inspection. GlobalProtect portal/gateway and app internals deferred to vendor docs. >